“INTERNET OF THINGS(IOT) BASED GAS DETECTION SYSTEM”

A PROJECT REPORT

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Submitted to

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Introduction:

Gas leakage is a serious safety hazard that can lead to accidents, explosions, and health issues. To address this problem, we have designed an IoT-based Gas Level Monitoring System using NodeMCU ESP8266, an MQ-2 gas sensor, and Blynk. The system continuously monitors the gas concentration and provides real-time alerts via LEDs, a buzzer, and a Blynk app notification when the gas level exceeds a safe threshold.

Components:

NodeMCU ESP8266: Microcontroller with WiFi support for IoT applications

MQ-2 Gas Sensor: Detects LPG, methane, and other combustible gases

Buzzer: Alerts users when the gas level is too high

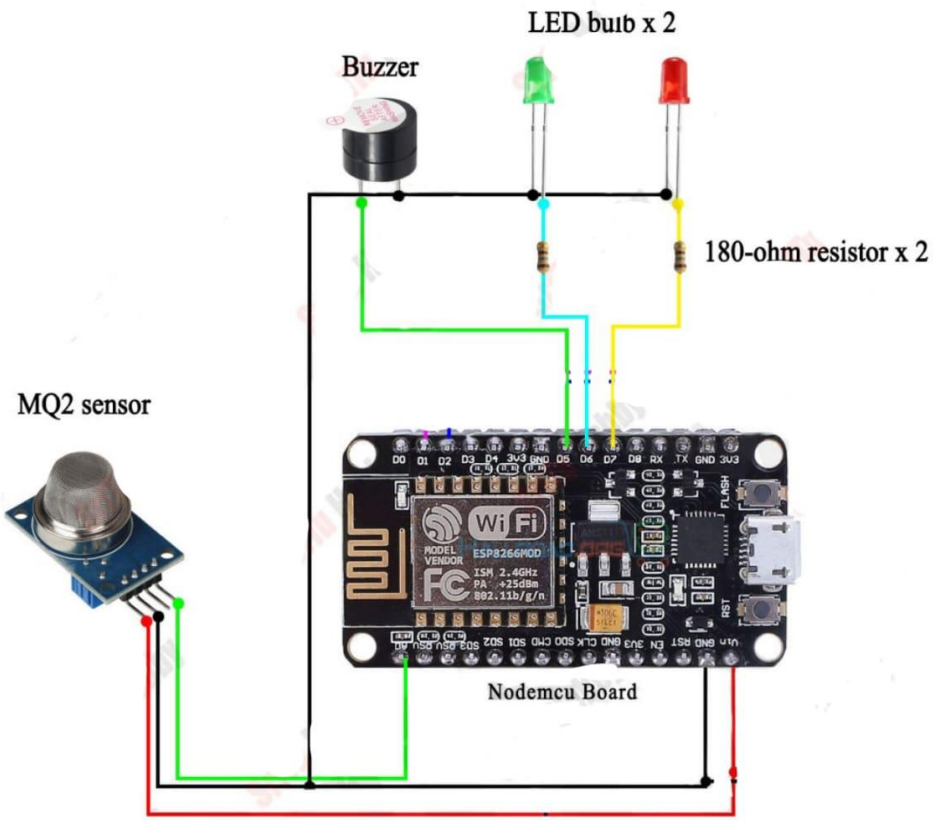
LEDs (Red & Green): Visual indicators for gas level status

Blynk App: Displays real-time gas levels and sends notifications

WiFi Connection: Enables cloud-based monitoring through the Blynk platform

Power Supply (5V): Provides necessary power to the components

Architecture of Project:

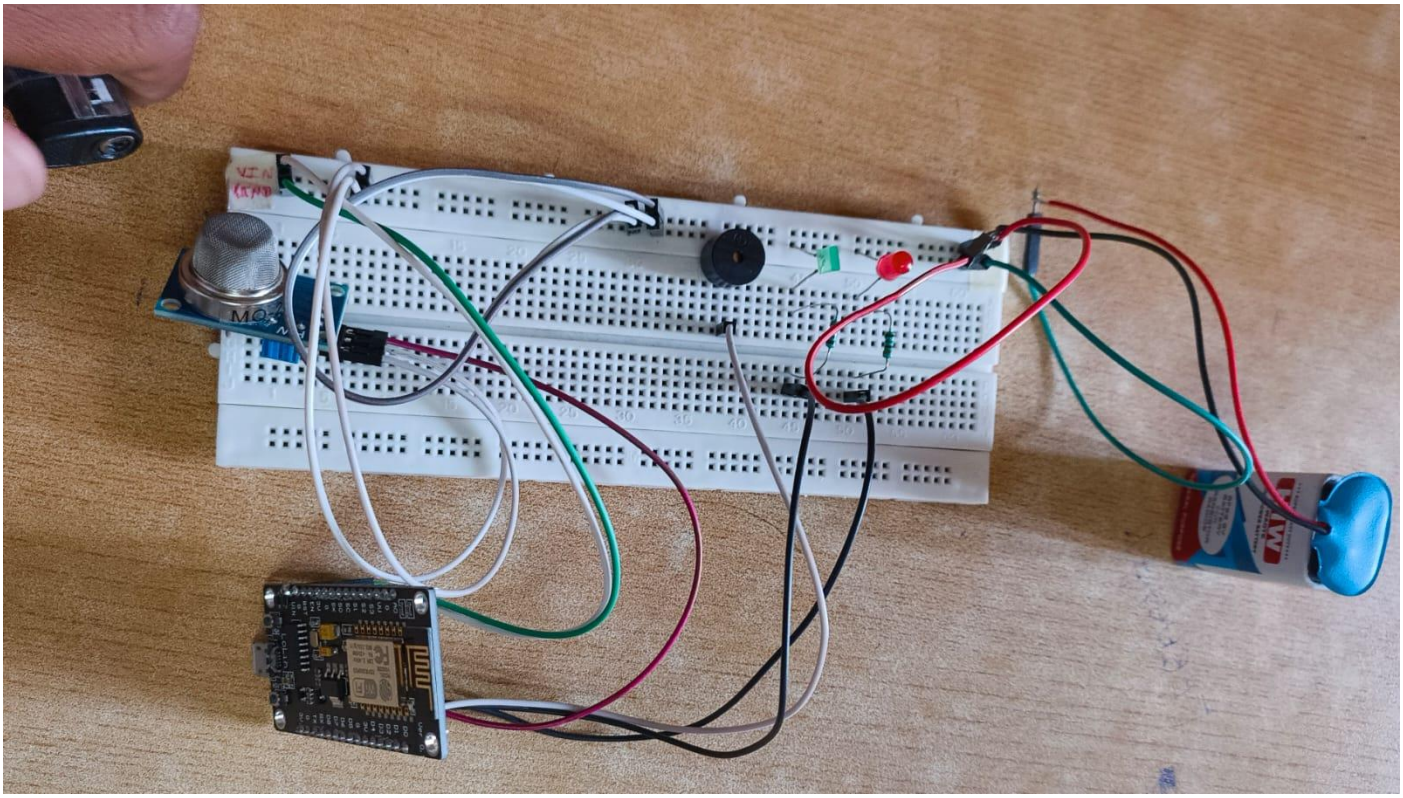


Working of the Project:

The system functions as follows:

1. The MQ-2 gas sensor continuously detects gas concentration and provides an analog output.
2. The NodeMCU ESP8266 reads the gas level from the sensor and sends the data to the Blynk app via WiFi.
3. The Blynk app (V0) displays the real-time gas level, and a button (V1) allows manual control.
4. If the gas level exceeds 300 (set threshold):
    - Buzzer activates 🔊
    - Red LED turns ON 🚨
    - Green LED turns OFF
    - Blynk sends a notification to alert the user 📱
5. If the gas level returns to normal:
    - Buzzer turns OFF
    - Red LED turns OFF
    - Green LED turns ON (indicating safe levels)

Conclusion:

The IoT Gas Level Monitoring System provides a real-time, smart, and efficient solution for gas leakage detection. By integrating WiFi, Blynk, and sensor technology, users receive instant alerts and can take immediate action to prevent hazards. This system is useful in homes, industries, and commercial spaces where gas leakage risks are high.

Picture of System: